

# 期末候選

1. 1.7.4 (a) + (b)

**1.7.4** The *girth* of  $G$  is the length of a shortest cycle in  $G$ ; if  $G$  has no cycles we define the girth of  $G$  to be infinite. Show that

- (a) a  $k$ -regular graph of girth four has at least  $2k$  vertices, and (up to isomorphism) there exists exactly one such graph on  $2k$  vertices;
- (b) a  $k$ -regular graph of girth five has at least  $k^2 + 1$  vertices.

2. 2.1.12

**2.1.12** A saturated hydrocarbon is a molecule  $C_mH_n$  in which every carbon atom has four bonds, every hydrogen atom has one bond, and no sequence of bonds forms a cycle. Show that, for every positive integer  $m$ ,  $C_mH_n$  can exist only if  $n = 2m + 2$ .

3. 2.2.3

**2.2.3** Show that if  $G$  is loopless and has exactly one spanning tree  $T$ , then  $G = T$ .

4. 2.2.11

**2.2.11** Show that if a graph  $G$  contains  $k$  edge-disjoint spanning trees then, for each partition  $(V_1, V_2, \dots, V_n)$  of  $V$ , the number of edges which have ends in different parts of the partition is at least  $k(n - 1)$ .

(Tutte, 1961 and Nash-Williams, 1961 have shown that this necessary condition for  $G$  to contain  $k$  edge-disjoint spanning trees is also sufficient.)

5. 2.3.2

**2.3.2** Show that a simple connected graph that has exactly two vertices which are not cut vertices is a path.

6. 2.4.4

2.4.4 Show that if  $e$  is an edge of  $K_n$ , then  $\tau(K_n - e) = (n-2)n^{n-3}$ .

7. 2.4.5 (c)

2.4.5 (a) Let  $H$  be a graph in which every two adjacent vertices are joined by  $k$  edges and let  $G$  be the underlying simple graph of  $H$ . Show that  $\tau(H) = k^{v-1}\tau(G)$ .

(b) Let  $H$  be the graph obtained from a graph  $G$  when each edge of  $G$  is replaced by a path of length  $k$ . Show that  $\tau(H) = k^{e-v+1}\tau(G)$ .

(c) Deduce from (b) that  $\tau(K_{2,n}) = n2^{n-1}$ .

8. prove that (找不到在課本哪裡)

$$\alpha(G) \geq \frac{|V(G)|}{\Delta(G) + 1}$$

$\alpha(G) \geq \frac{n}{1 + \Delta(G)}$

Let  $G$  be a connected graph of order  $n$  and  $\alpha(G)$  be the edge independence number (or matching number). Can you help me prove the inequality  $\alpha(G) \geq \frac{n}{1 + \Delta(G)}$ ?

 [https://math.stackexchange.com/questions/2601230/alpha\(G\)-geq-frac{n}{1-delta\(G\)}](https://math.stackexchange.com/questions/2601230/alpha(G)-geq-frac{n}{1-delta(G)})

9. prove that every maximal matching in a graph  $G$  has at least  $N$  edges (找不到在課本哪裡)

$$N = \alpha'(G)/2$$

10. 5.2.2 (a)

removed.

5.2.2 (a) Show that a bipartite graph  $G$  has a perfect matching if and only if  $|N(S)| \geq |S|$  for all  $S \subseteq V$ .

11. 5.2.6 (a)

- 5.2.6 (a) Prove the following generalisation of Hall's theorem (5.2): if  $G$  is a bipartite graph with bipartition  $(X, Y)$ , the number of edges in a maximum matching of  $G$  is

$$|X| - \max_{S \subseteq X} \{|S| - |N(S)|\}$$

(D. König, O. Ore)

12. 5.3.3

- 5.3.3** Show that a tree  $G$  has a perfect matching if and only if  $o(G - v) = 1$  for all  $v \in V$ .  
(V. Chungphaisan)